RWTH Aachen University
Institute for Data Science in
Mechanical Engineering

Bonus Point Assignment I

Reinforcement Learning and Learning-based Control

This document provides all necessary information concerning the first bonus point assignment. As this is an assignment for bonus points, some questions may extend beyond the scope of the lecture. The first bonus point assignment consists of three tasks. You can find all the task descriptions in the accompanying notebooks. Please follow the guidelines below while solving the tasks:

Formalia This is a voluntary assignment through which you can earn up to 4 bonus points for the final exam. Bonus points are only awarded if the final exam is passed. A failing grade cannot be improved using bonus points. The assignment must be completed in groups of 2 to 4 students. All group members must register in the same group via *"Groups Assignment 1"* on Moodle.

Dates Please note the following dates and deadlines, as they are strictly enforced:

1. Start of the assignment: 6^{th} May 2026, 00:01am (one minute past midnight)
2. Submission deadline: 17^{th} June 2026, 11:59pm (one minute before midnight)

Installation All tasks are to be completed using JupyterLab.
Detailed setup instructions are provided in the file `student_guide.md`. Ensure that your environment is correctly set up before starting the assignment. The notebooks are distributed via Moodle. Preserve the provided directory structure, as modifications may lead to errors during notebook execution. Save your work regularly and ensure that all group members have access to the latest version. While alternative tools (e.g., Google Colab or Visual Studio Code) may work, they frequently introduce compatibility issues during grading. Therefore, support is only provided for JupyterLab.

Tasks This bonus point assignment consists of three tasks, each provided as a separate notebook. You are required to complete and submit each notebook individually. The notebooks contain the complete and authoritative task description. All required libraries are already imported at the beginning of each notebook. Do not import additional libraries unless this is explicitly instructed in the task description. If the use of specific functionality (e.g., `torch.save()` from PyTorch) is required, this will be clearly stated. The notebooks use provided helper functions. These functions must not be modified. If you have questions, please use the bonus point assignments forum on Moodle. An overview of the tasks and their maximum points is given below:

Task	Title	Max points
01	Basketball Environment	10
02	Double-Q Learning	5
03	Off-Policy MC with Weighted Importance Sampling	5

Handling the Notebooks Your solutions will be checked automatically via unit tests. To ensure that grading works correctly, here are some essential guidelines:

- Submit your solutions as individual `.ipynb` notebooks. Each task must be uploaded separately to the corresponding Moodle assignment. You may name the notebook however you like.
- Do not rename any of the files for the submission.
- Do not delete or add any cells in the notebooks. If you accidentally modify the notebook structure, please download a fresh copy and transfer your solutions to this clean version.
- Stick to the class and function definitions given in the task description.
- Only place your answer in the highlighted cells. Each task includes a task description as well as several “student answer” and “checkpoint” cells. “Student answer” is for placing your answer, “checkpoint” helps you to debug.
- A “checkpoint” only provides information on whether the code runs without errors. However, errors cannot be ruled out this way. We therefore recommend that you carefully debug your code before submitting your work. It often helps to output values via `print()`.

Grading Submissions are evaluated automatically. The system checks for new submissions hourly and processes them in a queue. After grading, you will receive feedback through Moodle. This includes the number of points achieved and a detailed report indicating which parts were solved correctly and where points were lost. You may resubmit your solutions as often as you like until the submission deadline. We strongly recommend submitting early to receive feedback and to avoid longer waiting times due to increased system load close to the deadline. The total number of achievable points across all tasks is 20. Partial credit is awarded based on the number of passed unit tests. The points are converted into bonus points for the final exam as follows:

Points (BPA)	2	5	7	10	12	15	17	20
Bonus points (exam)	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0